The generic-degree spectrum of Keller counterexamples in dimension three

An audited, non-novel record

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Status, priority, and review notice. The degree-two exclusion is classical: Campbell proved the complex Galois case in 1973, followed by characteristic-zero algebraic proofs of Razar and Wright. Trushin's May 2026 preprint already contains the degree-two branch/singular-locus mechanism reproduced here. The minimum generic degree three and Gallagher's uniform every-degree weighted-lift theorem were public before this record. No novelty is claimed. This note is not peer reviewed. Exact computations check encoded identities only; they are evidence, not peer review or verification of the universal geometric arguments.

Abstract

We record the exact generic-degree spectrum of Keller counterexamples $\mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$. Generic degrees one and two cannot occur, by the classical theorem that a Keller map with Galois function-field extension is an automorphism. We also give a short independent proof of the quadratic case: a square-root generator would force a hypersurface into the codimension-two singular locus of a reduced hypersurface. Existing explicit weighted-lift maps realize every generic degree at least three. Hence the spectrum is exactly $\{3, 4, 5, \ldots\}$. Every cubic counterexample has geometric monodromy S_3. This note is an attributed research-program checkpoint, not a new result.

1 Statement and scope

Let k be a field of characteristic zero and let

$$F = (F_1, \ldots, F_n) : \mathbb{A}^n_k \longrightarrow \mathbb{A}^n_k$$

be polynomial with $\det JF \in k^\times$. Put

$$A = k[F_1, \ldots, F_n] \subset B = k[x_1, \ldots, x_n], \qquad K = \operatorname{Frac} A \subset L = \operatorname{Frac} B.$$

The *generic degree* is $[L : K]$.

Theorem 1 (Quadratic exclusion)**.** *If $[L : K] \leq 2$, then F is a polynomial automorphism. Thus a characteristic-zero Keller counterexample has neither generic degree one nor generic degree two.*

Theorem 2 (Exact complex spectrum)**.** *The generic degrees of Keller counterexamples $\mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$ are exactly*

$$\{3, 4, 5, \dots\}.$$

Every generic-degree-three counterexample has geometric monodromy S_3.

These statements concern function-field degree, not total polynomial degree $\max_i \deg F_i$. The latter is a separate problem. Vistoli's theorem in dimension three and total degree three gives the presently certified total-degree floor 4. Vistoli states that Moh and Sathaye had already obtained this result by an unpublished computer calculation; his 1999 paper gives the conceptual proof over an algebraically closed characteristic-zero field. Faithfully flat descent extends the floor to every characteristic-zero field.

2 The classical Galois proof

The Keller condition makes the F_i algebraically independent. A polynomial relation among them, after differentiation and multiplication by the inverse of JF, would have zero gradient; in characteristic zero it is constant. Thus F is dominant and L/K is finite and separable.

Campbell–Razar–Wright prove the following dimension-free statement.

Theorem 3 (Galois case [1, 2, 3])**.** *If $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ is Keller in characteristic zero and L/K is Galois, then $A = B$.*

The theorem requires neither properness nor surjectivity. A degree-one extension is trivially Galois, and every separable quadratic extension is Galois. This proves Theorem 1.

Remark 4. The hypothesis is that the original extension L/K is Galois. The existence of a Galois closure is not enough: every finite separable extension has one.

3 A self-contained quadratic argument

Over $\mathbb{C}$ the quadratic case has a short proof that exposes the codimension obstruction directly. Trushin's May 2026 preprint already contains the square-root and branch/singular-locus mechanism; the presentation below is a self-contained specialization, not a prior-art-independent proof.

Second proof of Theorem 1 over $\mathbb{C}$. Suppose $[L : K] = 2$. Since $A \simeq \mathbb{C}[y_1, \dots, y_n]$ is a unique factorization domain, the nontrivial square class can be represented by a nonconstant squarefree $h \in A$:

$$L = K(\sqrt{h}).$$

Set $g = \sqrt{h} \in L$. It satisfies

$$g^2 = h(F) \in B.$$

Thus g is integral over the integrally closed ring B, and consequently $g \in B$.

The polynomial g is nonconstant, so it has a zero. At every $p \in V(g)$,

$$0 = d(g^2)_p = d(h \circ F)_p = (dh)_{F(p)} \circ dF_p.$$

Since dF_p is invertible, dh vanishes at $F(p)$; also $h(F(p)) = 0$. Therefore

$$F(V(g)) \subseteq \operatorname{Sing} V(h). \tag{1}$$

The map F is etale and hence quasi-finite. Each irreducible hypersurface component of $V(g)$ has image closure of dimension $n-1$. But squarefreeness of h in characteristic zero makes $V(h)$ reduced, so it has no codimension-one singular component:

$$\dim \operatorname{Sing} V(h) \leq n-2. \tag{2}$$

Equations (1) and (2) contradict each other. □

This proof does not extend formally to a prime degree $p > 2$. Such an extension need not be Galois and need not be generated by a pth root that integrality places in the source polynomial ring.

4 Every generic degree at least three

We record an explicit existing weighted-lift family. Fix $d \geq 3$ and put

$$u = 1 + xy, \qquad \gamma = 1 - \frac{d}{d-1}xy + x^2 z.$$

Define $F_d = (A_d, B_d, C_d)$ by

$$A_d = \frac{(d-2)u + u^2 - (d-1)u^d\gamma^{d-2}}{(d-2)x^2},$$
$$B_d = \frac{(d-2) + 2u - du^{d-1}\gamma^{d-2}}{(d-2)x}, \qquad\qquad C_d = x\gamma. \tag{3}$$

For a direct divisibility check, put $v = xy$ and $\tau = x^2 z$. Both numerators are polynomials in v, τ; the numerator of B_d has zero constant term, whereas that of A_d has zero constant and zero v-linear term. Every remaining monomial is divisible by x and x^2, respectively. Hence (3) defines a polynomial map over $\mathbb{Q}$.

Set $w = u\gamma$, $P = B_d C_d$, and $Q = A_d C_d^2$. Elimination gives

$$w^d - w^2 + (d-2)Pw - (d-2)Q = 0. \tag{4}$$

After $U = (d-2)P$ and $V = -(d-2)Q$, the inverse equation becomes

$$T^d - T^2 + UT + V. \tag{5}$$

It is irreducible over $\mathbb{C}(U, V)$: as an element of $\mathbb{C}[U, V, T]$ it is primitive and linear in V, and any factor independent of V would have to divide its coefficient 1. Conversely, away from a proper denominator locus, a root w recovers the source through

$$\gamma = P - \frac{2w - dw^{d-1}}{d-2}, \quad x = \frac{C_d}{\gamma}, \quad u = \frac{w}{\gamma}, \quad y = \frac{u-1}{x}, \quad z = \frac{\gamma - 1 + \frac{d}{d-1}(u-1)}{x^2}. \tag{6}$$

Thus there is no branch introduced by denominator clearing, and the generic degree is exactly d.

For the Jacobian calculation, write

$$p_d(w) = \frac{2w - dw^{d-1}}{d-2}, \qquad q_d(w) = \frac{w^2 - (d-1)w^d}{d-2}.$$

The identity $q_d'(w) = wp_d'(w)$ collapses the 2×2 Jacobian after the target change

$$(A_d, B_d, C_d) \mapsto (B_d C_d, A_d C_d^2, C_d).$$

Comparison with the input changes $(x, y, z) \mapsto (x, xy, x^2 z) \mapsto (x, w, \gamma)$ gives

$$\det JF_d = 1. \tag{7}$$

It remains to certify noninjectivity. Let

$$s_d = \frac{4 - 2^d}{d - 2}.$$

Over the target $(A, B, C) = (s_d, s_d, 1)$, equation (4) has the distinct roots 1 and 2. The recovery denominators are

$$\gamma_1 = \frac{d + 2 - 2^d}{d - 2} \neq 0, \qquad \gamma_2 = 2^{d-1} \neq 0.$$

Equations (6) therefore give two distinct rational source points with the same image. So F_d is a counterexample of generic degree d. Together with Theorem 1, this proves Theorem 2.

5 Monodromy

A transitive subgroup of S_3 is either C_3 or S_3. If a separable cubic extension had geometric monodromy C_3, it would itself be Galois. Theorem 3 would then make the Keller map an automorphism. Hence every cubic counterexample has geometric monodromy S_3.

More generally, a counterexample cannot have regular monodromy. In a regular transitive action the point stabilizer is trivial, so the original extension is already Galois.

6 Verification and disclosure

The universal exclusion was checked against Campbell's complex-analytic covering/Hartogs argument, Wright's algebraic characteristic-zero argument, and Trushin's 2026 branch-divisor theorem. The singular-locus proof reproduced above is a specialization of Trushin's mechanism, not a fourth independent route. Companion SymPy and PARI/GP scripts regression-test (3), (7), and the explicit collision for $3 \leq d \leq 8$. Finite computation is not a proof for all d.

This record was produced with substantial AI assistance in theorem search, proof reconstruction, adversarial checking, exact computation, writing, and typesetting. It has not been peer reviewed and should not be represented as an independent discovery.

References

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